

Report Analysis on Image Enhancement Assignment

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In this experiment, a low-contrast city image was used to implement image enhancement using Histogram Equalization. The image was chosen because the details in the original image are not very clear due to the relatively low contrast. According to the lecture material, image enhancement is used to improve image quality so that information can be extracted more easily. Histogram processing is one of the pixel-based methods used for image enhancement.



Original Image

First, the original image was converted into grayscale. This was done so that the image could be analyzed based on its pixel intensity values. In a grayscale image, each pixel represents a different level of brightness. After converting the image, its histogram was displayed to see how the intensity values were distributed.



Grayscale Image

From the original histogram, most of the pixel values were concentrated within a limited range. This is consistent with the characteristics of a low-contrast image shown in the lecture material, where the histogram is concentrated in a smaller part of the available intensity range. Because the intensity values are not spread widely, the difference between darker and brighter areas of the image is not very strong. This can make some details in the city scene difficult to distinguish.

The next step was applying Histogram Equalization. Histogram Equalization is a gray-level transformation that spreads the transformed gray levels over the available intensity range. The lecture also explains that this method is fully automatic, data dependent, and usually enhances contrast. In the implementation, the `cv2.equalizeHist()` function was used to perform this process on the grayscale image.



Histogram Equalized Image

After Histogram Equalization was applied, the resulting image showed a clearer difference between darker and brighter areas. Some details that were less noticeable in the original image became easier to see. The histogram also changed compared to the original histogram. Instead of being concentrated mainly in a limited intensity range, the pixel values were distributed across a wider range. This indicates that the contrast of the image was increased.

The comparison between the original and enhanced images shows that Histogram Equalization can be useful for improving a low-contrast image. However, the result is not necessarily perfect in every area of the image. Since Histogram Equalization changes the intensity distribution of the entire image, some parts can become more pronounced than others. The lecture material also notes that there are cases where Histogram Equalization can produce worse results, so the enhanced image still needs to be checked visually rather than only relying on the histogram.

Overall, the experiment successfully demonstrated the use of Histogram Equalization for image enhancement. The original city image had relatively low contrast, while the enhanced image had a wider distribution of pixel intensities and clearer visual details. The histogram comparison helped show how the pixel distribution changed after processing. Therefore, Histogram Equalization can be used as a simple pixel-based method to improve the contrast of a low-contrast image.